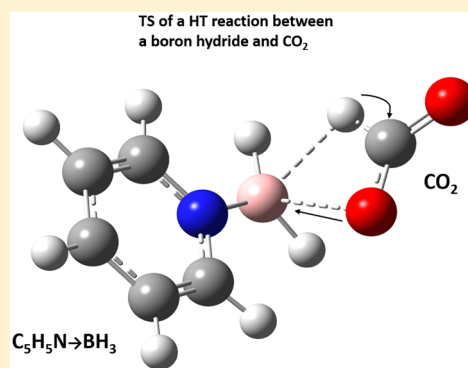


# Predicting Hydride Donor Strength via Quantum Chemical Calculations of Hydride Transfer Activation Free Energy

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## S Supporting Information

**ABSTRACT:** We propose a method to approximate the kinetic properties of hydride donor species by relating the nucleophilicity ( $N$ ) of a hydride to the activation free energy  $\Delta G^\ddagger$  of its corresponding hydride transfer reaction.  $N$  is a kinetic parameter related to the hydride transfer rate constant that quantifies a nucleophilic hydridic species' tendency to donate. Our method estimates  $N$  using quantum chemical calculations to compute  $\Delta G^\ddagger$  for hydride transfers from hydride donors to  $\text{CO}_2$  in solution. A linear correlation for each class of hydrides is then established between experimentally determined  $N$  values and the computationally predicted  $\Delta G^\ddagger$ ; this relationship can then be used to predict nucleophilicity for different hydride donors within each class. This approach is employed to determine  $N$  for four different classes of hydride donors: two organic (carbon-based and benzimidazole-based) and two inorganic (boron and silicon) hydride classes. We argue that silicon and boron hydrides are driven by the formation of the more stable Si–O or B–O bond. In contrast, the carbon-based hydrides considered herein are driven by the stability acquired upon rearomatization, a feature making these species of particular interest, because they both exhibit catalytic behavior and can be recycled.



## 1. INTRODUCTION

More than 30 billion tons of carbon dioxide ( $\text{CO}_2$ ) are released into the atmosphere annually, but only about half of this  $\text{CO}_2$  is naturally recycled by photosynthesis,<sup>1</sup> resulting in a net accumulation of  $\text{CO}_2$  in the atmosphere. Accordingly, there has been a growing interest in developing economical ways to chemically reduce the accumulating  $\text{CO}_2$  to fuels such as methanol in order to close the carbon cycle.<sup>2–4</sup> Several groups have successfully converted  $\text{CO}_2$  to methanol via hydride transfer (HT) pathways. For instance, Ying and co-workers<sup>5</sup> showed that  $\text{CO}_2$  was reduced to methanol by diphenylsilane and an  $n$ -heterocyclic carbene catalyst,<sup>6</sup> while Fontaine and co-workers demonstrated the hydroboration of  $\text{CO}_2$  to methanol<sup>7</sup> with the aid of frustrated Lewis pairs.<sup>8–10</sup> While silanes and boranes are effective in reducing  $\text{CO}_2$ , they are not catalytic but are instead stoichiometrically consumed in the reaction. In contrast, carbon-based hydrides, specifically dihydropyridines, are renewable hydrides capable of catalytically reducing  $\text{CO}_2$ .<sup>11–16</sup>

Efforts to quantify the hydride donor abilities of hydrides to effect reductions have been reported and can significantly aid the selection of appropriate hydride donors for  $\text{CO}_2$  reduction. For example, Mayr and co-workers have discussed a method to quantify the kinetic hydride nucleophilicity ( $N$ ) of various

hydrides including boranes, silanes, and carbon-based hydrides.<sup>17</sup> Specifically, these authors proposed that, as shown in the definition relation eq 1, the logarithm of the (dimensionless numerical value of the) rate constant ( $\text{M}^{-1} \text{s}^{-1}$ ) of a HT reaction at 20 °C is equal to the sum of  $N$  and the electrophilicity ( $E$ ) of the hydride acceptor, multiplied by a sensitivity parameter ( $s_N$ ) that depends only on the nature of the nucleophile:

$$\log k_{20^\circ\text{C}} = s_N(N + E) \quad (1)$$

By measuring the rates of HT from a hydride donor (nucleophile) to various hydride acceptors (electrophiles) with known electrophilicities  $E$ , the desired  $N$  and  $s_N$  values can be determined for the hydride donor of interest from eq 1 via linear regression. The sensitivity parameter  $s_N$  is typically within the range 0.5–1.2,<sup>18</sup> and is approximately constant within each class of hydride donors (*vide infra*);  $N$  typically spans the range from 0 to 15, with larger values indicating stronger nucleophiles. We emphasize that, as a kinetic parameter,  $N$  is distinct from the often employed thermodynamic hydricity parameter.<sup>19</sup>  $N$  and the hydricity of a molecule thus provide

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different measures for quantifying the strength of the hydride in HT reactions.<sup>20</sup>

Mayr and co-workers showed that, when tabulated values of  $s_N$ ,  $N$ , and  $E$  are available, eq 1 provides a reasonable approximation for the HT rate constant.<sup>18</sup> Of the 109 experiments conducted on a series of hydride donors and acceptors, only four rate constants predicted via eq 1 deviated from experimentally determined values by a factor larger than 50, i.e., by less than a factor of 2 in the log of  $k$ .<sup>17</sup> Thus, the  $N$  values predicted by eq 1 provide a robust scale<sup>21</sup> to quantify and predict the hydride donor abilities of various classes of organic and other hydrides.<sup>22</sup>

In this work, we demonstrate that computational chemistry can be utilized as a tool to accurately predict nucleophilicity  $N$  values for various classes of hydrides that include silanes, boranes, and carbon-based hydrides. Together with eq 1, this enables the prediction of the compounds' reactivity before their synthesis, thereby providing a rapid screening process to guide the experimental development of hydride donors.

Several groups have introduced approaches to computationally approximate both  $N$  and  $E$  values. While these approaches have proven to provide good approximations, they simultaneously utilize numerous thermodynamic and other data—such as ionization potential, electron affinity, chemical hardness, and electrochemical potential—as descriptors to predict the  $N$  and  $E$  kinetic parameters.<sup>23,24</sup> In contrast, we use a more direct approach to predict the  $N$  values of hydrides, requiring only a single parameter—the activation free energy ( $\Delta G^\ddagger$ )—and demonstrate its accuracy by recovering the Mayr group's experimentally measured  $N$  values ( $E$  values are discussed later). To this end, we first establish a fundamental linear relationship between the quantum chemically computed activation free energies  $\Delta G^\ddagger$  for hydride transfer and experimentally measured  $N$  values; then, using the computed  $\Delta G^\ddagger$  for hydride donors of interest, we apply this relationship to predict the donors'  $N$  values. Our approach involving eq 1 is analogous to the experimental determination of  $N$  requiring HT rate constant measurement; our approach effectively replaces this measurement with the calculation of  $\Delta G^\ddagger$ . An additional feature of the present effort is that we establish correlations of the  $\Delta G^\ddagger$ – $N$  relations with the stabilities of the HT reaction products, which differ between two different HT reaction types.

We envision that this approach will aid the development of competent hydride donor catalysts not only for the reduction of  $\text{CO}_2$  to usable fuels but also for hydride donor catalysts for organic synthesis in general.

## 2. COMPUTATIONAL METHODS

Geometries of all reactants, products, and transition states (TSs) were computed using density functional theory based on the M06 density functional<sup>25</sup> combined with the 6-31+G(d,p) basis set as implemented in Gaussian 09.<sup>26,27</sup> Proper treatment of the solvent is necessary to correctly describe reactions in solution involving polar TSs. An implicit polarized continuum solvation model (CPCM) was used to simulate solvation effects for the reactions considered, which all involve hydride transfers and thus polar TSs.<sup>28,29</sup> We need to correctly calculate the reactions' TS in a CPCM-simulated solvent environment similar to Mayr and co-workers' experiments;<sup>17</sup> accordingly, CPCM dichloromethane solvent was utilized for all reactions. This method reproduced the HT barriers of 10 hydrides calculated with the wB97XD/6-311+G(d,p) level of theory

within 2.1 kcal/mol, as shown in section C of the Supporting Information.<sup>30,31</sup>

Vibrational force constants at the M06/6-31+G(d,p) level of theory were calculated to (1) verify that the reactant and product structures have only positive vibrational frequencies, (2) confirm that each TS has only one imaginary mode and that it connects the desired reactant and product structures via intrinsic reaction coordinate calculations,<sup>32</sup> and (3) compute entropies, zero-point energies, and thermal corrections included in the reported free energies at 298 K.

For the calculated activation and reaction enthalpies, entropies, and free energies, the reference state is defined as the separated reactants in solution, as is appropriate for solution-phase bimolecular reactions.<sup>33</sup>

Activation entropies  $\Delta S^\ddagger$  for the HT reactions require special attention. Commonly employed entropy evaluations within the rigid rotor, harmonic oscillator, and ideal gas approximations generally overestimate the entropic penalty for solution phase reactions because ideal gas partition functions do not explicitly account for hindered translations, rotations, or vibrations of the solvated solute molecule. We follow two different methods to correct these errors, both of which yield equally accurate recovery of  $N$  values. The first approach is to omit the contributions of translational gas-phase entropies as proposed by Morokuma and co-workers.<sup>34</sup> The second approach<sup>12</sup> involves adding the experimentally determined entropic contributions  $-T\Delta S^\ddagger_{\text{exp}}$  to the calculated activation enthalpy  $\Delta H^\ddagger$ .<sup>35</sup> This provides a better approximation to the solvated reactions  $T\Delta S^\ddagger$  by removing or “quenching” the translational entropy term contributions to the free energy.

We now detail this second approach. Srinivasan et al. conducted experiments on HT between 1-benzyl-1,4-dihydro-nicotineamide (hydride donor) and  $\Delta^1$ -pyrroline-2-carboxylic acid (hydride acceptor) in water, finding that this reaction has an entropy of activation of  $-7.6$  eu, which translates to a  $-T\Delta S^\ddagger_{\text{exp}}$  value of 2.3 kcal/mol at 298 K.<sup>35</sup> We will use this value for HT reactions in the second approach, which makes two assumptions: (1) the reaction studied by Srinivasan et al. behaves similarly to  $\text{CO}_2$  reduction by carbon-based hydrides, so we expect similar entropic contributions to  $\Delta G^\ddagger$  for the two systems,<sup>12</sup> and (2) regarding all of the HT classes examined here, most of the molecules we consider are of similar size and mass and therefore are expected to exhibit, for all classes, similar entropic contributions from translational, rotational, and vibrational modes for all classes. The latter assumption is supported by calculations of  $T\Delta S^\ddagger$  at the M06 level of theory,<sup>25</sup> which, despite exaggerating the entropic contribution, predict  $T\Delta S^\ddagger$  for boron hydrides to  $\text{CO}_2 \sim 1$  kcal/mol smaller than  $T\Delta S^\ddagger$  of carbon-based, benzimidazole-based, and silicon-based hydrides, an amount well within the error for the calculated enthalpic contribution to  $\Delta G^\ddagger$ .

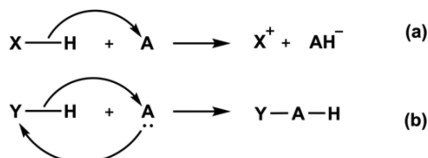
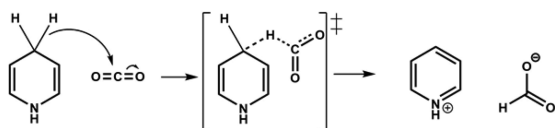
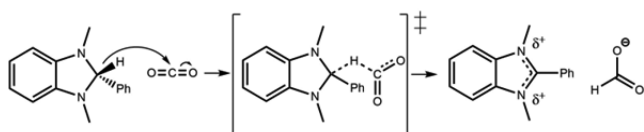
Both methods for estimating the HT reaction activation entropy provide similar results; here we will present those of the first method and report the second method's results in the Supporting Information.

## 3. RESULTS AND DISCUSSION

**3.1. Calculation of  $N$  Values.** We first indicate how the nucleophilicity kinetic parameter  $N$  can be calculated directly from the activation free energy  $\Delta G^\ddagger$  of the hydride transfer. Using simple algebraic manipulations of the Mayr group's model, we will then attempt to recover their experimentally inferred  $N$  values and, in addition, predict  $N$  values for hydride

**Table 1.** Ranges of Experimentally Derived  $s_N$  Values for Different Classes of Hydrides<sup>17</sup>

| class         | range of $s_N$ |
|---------------|----------------|
| carbon        | 0.80–1.10      |
| benzimidazole | 0.70–0.72      |
| boron         | 0.67–0.81      |
| silicon       | 0.58–0.79      |

**Figure 1.** Reactions with different hydride transfer behavior. The schematic reaction type (a) exemplifies the reaction character trends of carbon-based and benzimidazole-based hydride classes; after hydride donation, the product species are stable as an anion and a cation. The schematic reaction path type (b) shows the contrasting behavior of silicon- and boron-hydride groups, where the product is more stable as a single complex molecule.**Scheme 1.** Hydride Transfer Reaction Mechanism between a Carbon-Based Hydride Donor and CO<sub>2</sub>**Scheme 2.** Hydride Transfer Reaction Mechanism between a Benzimidazole-Based Hydride Donor and CO<sub>2</sub>

molecules whose values are not yet known. Our terms “recovery” and “prediction” respectively refer to obtaining  $N$  values for molecules that already have or do not yet have experimentally determined  $N$  values.

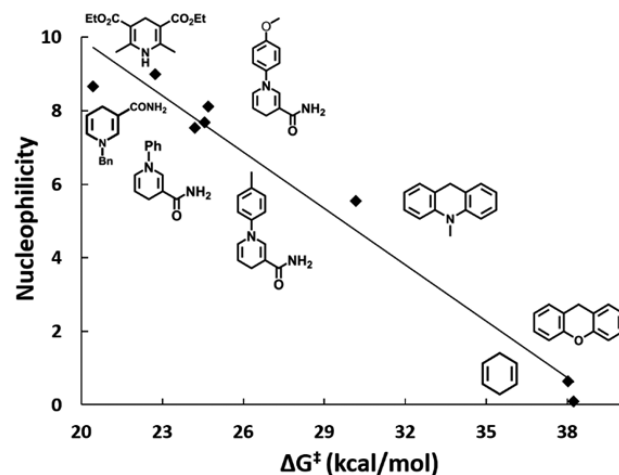
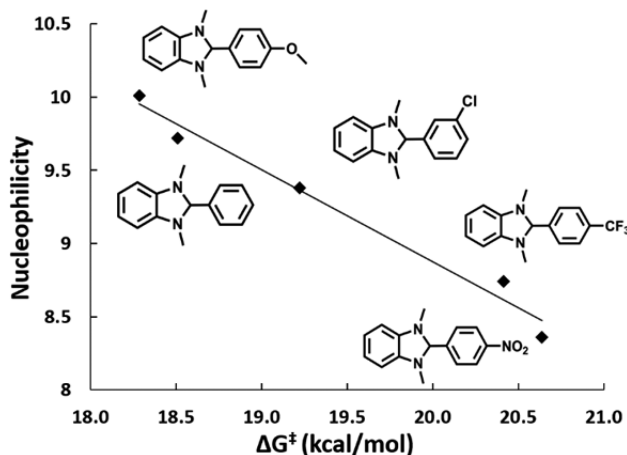
A simple algebraic manipulation shows that  $N$  and  $\Delta G^\ddagger$  are linearly related to each other (i.e.,  $N \propto \Delta G^\ddagger$ ) when the proportionality coefficient  $s_N$  is approximately constant within a given class of hydride donors being analyzed. To demonstrate this, we combine the well-known transition state theory-based equation<sup>36</sup> for the exponential dependence of the rate of reaction on the activation free energy

$$k = c_0 \frac{k_B T}{h} \exp(-\Delta G^\ddagger / RT) \quad (2)$$

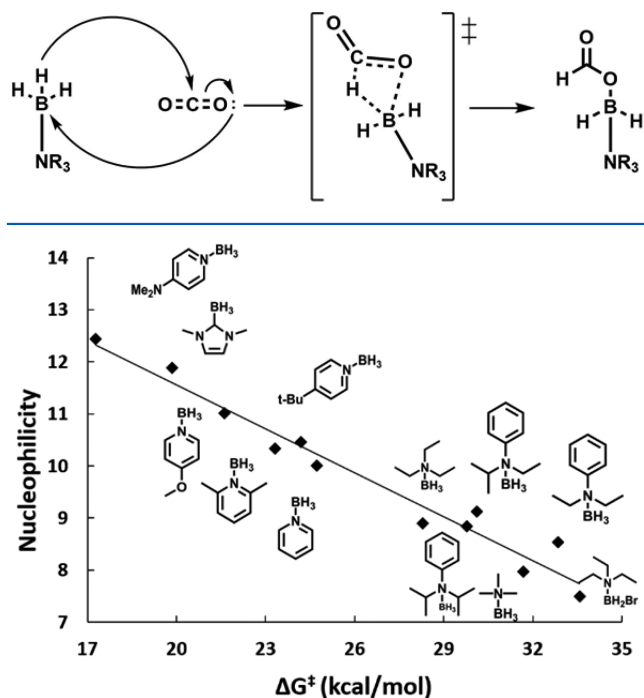
where  $k_B$  and  $h$  are the Boltzmann and Planck constants and  $c_0$  is a factor to guarantee units of  $\text{M}^{-1} \text{s}^{-1}$  for the dimensions of  $k$ , with the Mayr group's definition of nucleophilicity, eq 1, to yield eq 3.

$$N = -\frac{1}{2.3RTs_N} \Delta G^\ddagger + \frac{1}{s_N} \log_{10} \left( u c_0 \frac{k_B T}{h} \right) - E \quad (3)$$

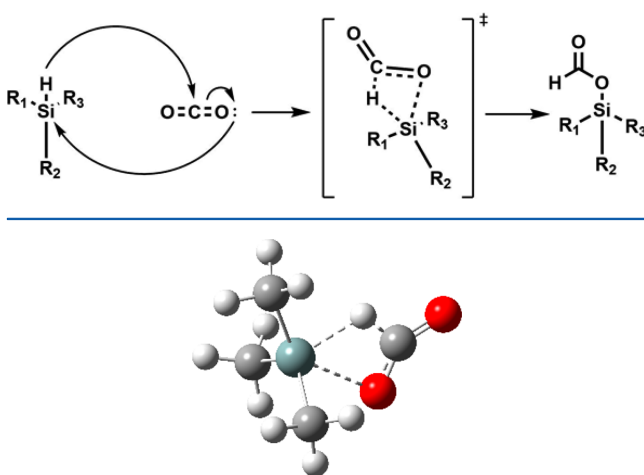
Equation 3 connects  $N$  and  $\Delta G^\ddagger$  and relates three unknowns:  $\Delta G^\ddagger$ ,  $N$ , and  $E$ .  $u$  has units of  $\text{M} \cdot \text{s}$ ; hence, the argument of the log is dimensionless. All of the reactions are

**Figure 2.** Nucleophilicity  $N$  vs activation free energy of reaction  $\Delta G^\ddagger$  for carbon-based hydride donors with CO<sub>2</sub> as the hydride acceptor in dichloromethane solvent. Experimental data are labeled as diamonds, while the line represents calculated  $N$  values based on eq 3,  $N = -0.51 \Delta G^\ddagger + 20.2$ , and computed  $\Delta G^\ddagger$  for hydride transfers. Effective  $s_N$  and  $E$  values of 1.44 and  $-11.3$  are obtained. The mean absolute deviation (MAD) of the differences between experimental and calculated nucleophilicity values is 0.26. A paired  $t$  test on each set of the experimental and recovered  $N$  values to determine the effectiveness of the model results in a 95% confidence interval of  $(-0.52, 0.52)$  and a  $p$ -value of 1.0; a  $p$ -value greater than 0.05 indicates that no significant difference exists between recovered and experimentally inferred nucleophilicity values, which is validated by the inclusion of 0 in the 95% confidence interval.<sup>42</sup>**Figure 3.** Nucleophilicity  $N$  vs activation free energy of reaction  $\Delta G^\ddagger$  for benzimidazole-based hydride donors with CO<sub>2</sub> as the hydride acceptor in dichloromethane solution. Experimental data are labeled as diamonds, while the line represents calculated  $N$  values based on eq 3,  $N = -0.63 \Delta G^\ddagger + 21.4$ , and computed  $\Delta G^\ddagger$  for hydride transfers. Effective  $s_N$  and  $E$  values of 1.17 and  $-10.5$  are obtained.  $s_{N,\text{eff}}$  is significantly outside the specified range noted in Table 1. As is to be discussed in the General Trends section, we believe this results from the differing nature of the analysis between this work and those of Mayr and co-workers. The MAD between the experimental and recovered nucleophilicity values is 0.03 with a 95% confidence interval of  $(-0.13, 0.12)$  and a  $p$ -value of 0.98. The confidence interval and  $p$ -value indicate that there is no significant difference between the experimental and recovered nucleophilicity values.

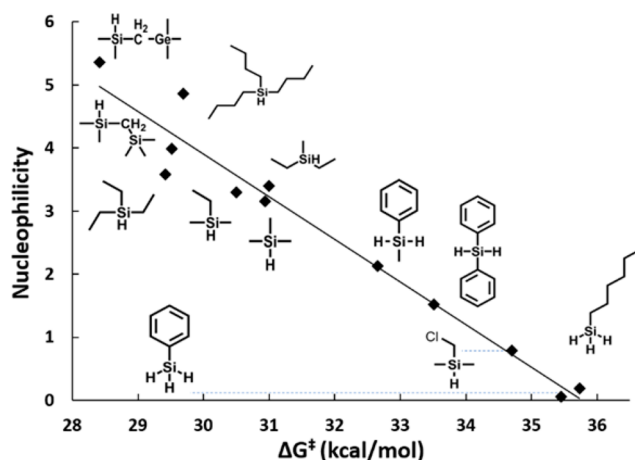
considered at the constant temperature  $T = 298$  K. The strictly linear relationship between  $N$  and the computed  $\Delta G^\ddagger$  only

Scheme 3. Hydride Transfer Reaction Mechanism between a Boron-Based Hydride Donor and CO<sub>2</sub>

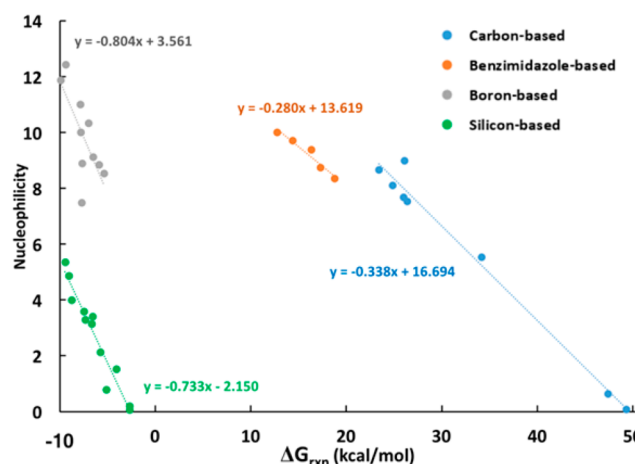
**Figure 4.** Nucleophilicity  $N$  vs activation energy of reaction  $\Delta G^\ddagger$  for boron-based hydride donors with CO<sub>2</sub> as the hydride acceptor in dichloromethane. Experimental data are labeled as diamonds, while the line represents calculated  $N$  values based on eq 3,  $N = -0.28\Delta G^\ddagger + 17.2$ . Effective  $s_N$  and  $E$  values of 2.6 and  $-12.3$  are obtained. Note that the effective  $s_N$  value is outside the range of  $s_N$  values for boron hydrides reported in Table 1. As is to be discussed in the General Trends section, we believe this results from the differing nature of the analysis between this work and those of Mayr and co-workers. The MAD between the experimental and recovered nucleophilicity values is 0.12. A paired  $t$  test predicts a 95% confidence interval of  $(-0.19, 0.19)$  and a  $p$ -value of 1.0, indicating that the difference between the recovered and experimental nucleophilicity values is insignificant.

Scheme 4. Hydride Transfer Reaction Mechanism between a Silicon-Based Hydride Donor and CO<sub>2</sub>

**Figure 5.** Structure of the transition state of the hydride transfer reaction between HSi(CH<sub>3</sub>)<sub>3</sub> and CO<sub>2</sub>. A silicon–oxygen bond begins to form as the hydride moiety dissociates from silicon and approaches carbon.

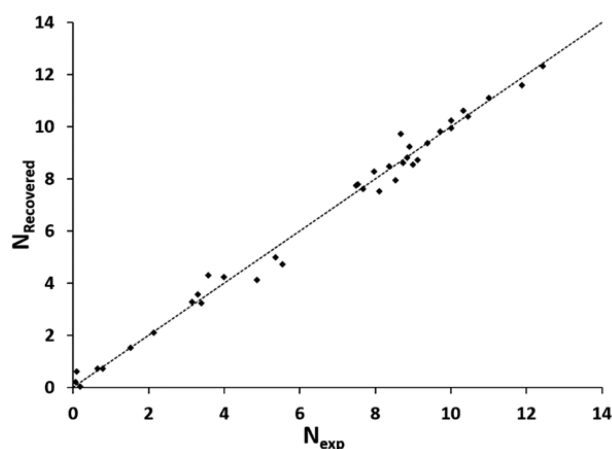


**Figure 6.** Nucleophilicity  $N$  vs activation free energy of reaction  $\Delta G^\ddagger$  for silicon-based hydride donors with CO<sub>2</sub> as the hydride acceptor in dichloromethane. Experimental data are labeled as diamonds, whereas the line represents computationally calculated  $N$  values. The line  $N = -0.68\Delta G^\ddagger + 23.9$  is shown, based on eq 3. Effective  $s_N$  and  $E$  values of 1.08 and  $-12.1$  are obtained. Note that this effective  $s_N$  value is outside the range of  $s_N$  values for silicon hydrides reported in Table 1, and this discrepancy will be discussed in the General Trends section. The MAD of the difference between experimental and recovered nucleophilicity values is 0.18. A paired  $t$  test between experimental and recovered  $N$  values yielded a 95% confidence interval of  $(-0.23, 0.23)$  with a  $p$ -value of 0.99.



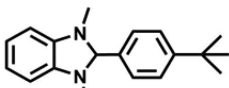
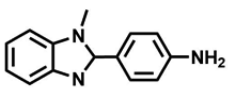
**Figure 7.** Linear relationships between the thermodynamic and kinetic properties of hydride transfer.  $N$ , which is related to the free energy barrier, scales linearly with  $\Delta G_{\text{rxn}}$  of HT reactions for carbon- (blue), benzimidazole- (orange), boron- (gray), and silicon-based (green) hydride donors. The MAD between all experimental  $N$  values and those predicted using this method was found to be 0.41. The  $p$ -value for the paired  $t$  test between predicted and experimental  $N$  is 0.99, with a 95% confidence interval of  $(-0.22, 0.22)$ . The linear fits result in  $R^2 > 0.92$  for silicon-, carbon-, and benzimidazole-based hydrides and  $R^2 = 0.52$  for B-based hydrides, where the two outliers lie far off the linear fit.

holds for a constant electrophilicity value  $E$ ; we ensure this constancy by selecting CO<sub>2</sub> as the electrophile hydride acceptor for all reactions we consider. The last two terms in eq 3 can be combined into an intercept  $N_0$  term such that the relationship takes the linear form shown in eq 4.



**Figure 8.** Recovered vs experimental nucleophilicities  $N_{\text{exp}}$ .<sup>17</sup> The 45° dashed line represents perfect agreement between both values. Because  $N$  is logarithmically scaled to the rate constant  $k$ , a difference of 1 nucleophilicity unit represents a deviation of a factor of 10 in the rate constant at 298 K. Of course, some errors may result from experimental errors in the reported rates of hydride transfers.

**Table 2. Predictions of Previously Undetermined  $N$  Values for Hydride Donors Belonging to the Four Hydride Classes Defined in This Work<sup>a</sup>**

| Hydride Molecule                                                                    | $N$ -value |
|-------------------------------------------------------------------------------------|------------|
|   | 10.9       |
|  | 9.9        |
|  | 10.0       |
|  | 10.6       |
|  | 9.0        |
|  | 9.4        |
|  | 3.1        |
|  | 1.4        |

<sup>a</sup>See the Supporting Information for a comparison of results between approach 1 and approach 2 for the estimation of the HT reaction activation entropy.

$$N = -\frac{1}{2.3RTs_N} \Delta G^\ddagger + N_0 \quad (4)$$

Our results reported below show that the resulting nucleophilicity is indeed linearly dependent on the HT activation free energy barrier for each class of hydrides.<sup>37</sup> Mayr and co-workers have effectively already demonstrated this because  $\ln k$  is linear in  $\Delta G^\ddagger$ ; with the novel feature that we calculate  $\Delta G^\ddagger$ , we show here that the linear relationship between  $N$  and  $\Delta G^\ddagger$  holds for a given fixed acceptor—separately for each class, and, further, we will provide some mechanistic interpretation for the  $s_N$  values in the relation.

While other hydride acceptors could have been selected for the present analysis, we have chosen  $\text{CO}_2$  for the reasons indicated in the Introduction: significant interest exists for chemical reductions of  $\text{CO}_2$  via HTs,<sup>2</sup> and modeling HT to  $\text{CO}_2$  reactions provides a direct prediction of  $\text{CO}_2$ 's reactivity toward different hydrides. In fact, our own ultimate goal is to discover HT catalysts that both convert  $\text{CO}_2$  into methanol and provide a robust process to close the carbon cycle. An additional attraction of the choice of  $\text{CO}_2$  is its small size that makes it convenient for performing large numbers of quantum chemical calculations to screen hydride donors, facilitating the employment of more accurate and computationally demanding electronic structure methods.

We will consider four different classes of hydrides in our analysis, the first two of which, comprising carbon-based and benzimidazole-based hydrides, involve organic hydrides. All carbon-based hydrides we discuss become aromatic in nature upon hydride transfer. Although carbon-based hydrides exist that are nonaromatic upon hydride transfer (e.g., 3-methyl-1,4-pentadiene),<sup>17</sup> we have omitted them from the present analysis due to their extremely weak nucleophilic capabilities. The benzimidazole-based hydrides are considered as a separate class from carbon-based hydrides due to a qualitative difference in driving force for their reactivity: as will be discussed in more detail later, the latter hydrides stabilize by regaining aromaticity after a HT, whereas the former stabilize by resonating  $\pi$ -bonds. Finally, the other two, inorganic, classes involve boron and silicon inorganic hydrides, which are distinguished by the atom to which the hydridic hydrogen is bonded.

Our analysis deals straightforwardly with the HT reaction electrophile, since in all four models  $\text{CO}_2$  acts as the hydride acceptor and thus the hydride acceptor's electrophilicity in all four models must be that of  $\text{CO}_2$ . Equation 4 for the relation between  $N$  and  $\Delta G^\ddagger$  dictates that the  $y$ -intercept  $N_0$  includes the electrophilicity of the hydride acceptor. Equations 3 and 4 can be combined to solve for  $E_{\text{eff}}$  in terms of the intercept and the effective  $s_N$  value.

$$E_{\text{eff}} = \frac{1}{s_N} \log_{10} \left( \frac{k_B T}{h} \right) - N_0 \quad (5)$$

$E_{\text{eff}}$  values are calculated for each class of hydrides considered and span a small range of values (−12.3, −10.5), indicating that  $\text{CO}_2$  has a relatively similar  $E$  value in all cases considered (the range shifts to (−14.9, −14.1) when the second method for estimating  $T\Delta S^\ddagger$  is used; see section B of the Supporting Information).

**3.2. Sensitivity Parameter  $s_N$ .** While the hydride acceptor's electrophilicity in eq 3 is taken to be constant—here required with the same electrophile ( $\text{CO}_2$ ) considered for all HT reactions—the coefficient  $s_N$  is not necessarily constant. This sensitivity parameter  $s_N$  is a property of each nucleophile

and in the Mayr analysis can range from 0.5 to 1.2.<sup>18</sup> Thus, each nucleophile may have a distinct sensitivity depending on its reactivity with the hydride acceptor, so that eq 3 is not automatically linear. However, in the present work, we will assume that, within each class of hydrides,  $s_N$  is constant. This assumption is supported by the resulting accurate predictions of  $N$ . It also has some support from the demonstration by Horn et al. for silanes that correct  $N$  values could be determined from the experiments of Chojnowski and co-workers by assuming a constant  $s_N$  value of 0.75.<sup>17,38</sup> Our approach involves making a similar assumption where the sensitivity parameter of each class can be approximated by a small range of values. If  $s_N$  does not change significantly within a class, the relation between  $N$  and  $\Delta G^\ddagger$  should be approximately linear. Table 1 indicates the resulting parameter range we have found for each of the classes.

Thus, in our analysis, it is the sensitivity parameter that varies between the HT reaction classes. In the work of Mayr and co-workers,  $s_N$  is described as a measure of the sensitivity of a nucleophile to changes in the electrophile, with a reference  $s_N$  value of 1 for 2-methyl-1-pentene.<sup>18</sup> These authors made the empirical observation that nucleophiles that are more sensitive to the electrophile have larger  $s_N$  values. Since we do not vary the electrophile in the present study, we follow here a different line of reasoning in attempting to provide an explicit physical interpretation of  $s_N$  before entering into the detailed numerical analysis; in particular, we focus instead on the nucleophile. In this connection, two facts regarding  $s_N$  are considered: (a)  $s_N$  depends only on the nucleophilic molecule, and (b) each family of hydrides exhibits a small range of  $s_N$  values. We reason as follows. Upon hydride abstraction, a positive charge is localized on the oxidized hydride donor, and each class of hydrides will stabilize the positive charge differently. Within a class, we can expect that there is a correlation between the increased reaction rate constant—proportional to  $s_N$ —and the increased thermodynamic stabilization of the reaction product. We then anticipate that the variation of  $s_N$  values between the four classes correlates with the stabilizing forces of the HT reaction product.

To examine the anticipated correlation between such stabilizing forces and  $s_N$ , we now consider two reaction types in Figure 1 that occur throughout our calculations below, where  $X$  and  $Y$  are hydride donors and  $A$  is the hydride acceptor. Of the four reaction classes we examine, all of the carbon-based and benzimidazole-based hydrides follow reaction type (a) in Figure 1a, whereas the two inorganic classes, the silanes and boranes, follow reaction type (b) in Figure 1b.

Although carbon-based and benzimidazole-based hydrides follow the same reaction type (a), their stabilizing forces in the HT reactions differ. Carbon-based hydrides regain their aromaticity upon hydride donation (Scheme 1), whereas benzimidazole-based hydrides stabilize the resulting positive charge of HT by resonating  $\pi$ -bonds between N and C atoms (Scheme 2). Rearomatization is considerably more stabilizing than simple resonance,<sup>39</sup> rationalizing the larger  $s_N$  value of carbon-based hydrides relative to benzimidazole-based hydrides listed in Table 1.

For the inorganic hydrides, we will discuss the rationale for their  $s_N$  values in terms of the more specific case of reducing  $\text{CO}_2$  to formate. Silicon-based and boron-based hydrides both follow reaction type (b) in Figure 1, and they both stabilize their positive charge by forming a B or Si bond to oxygen in a combination reaction with formate. In order to quantify the stabilizing forces in these two inorganic classes, the difference in

bond energies of the reactant and product is calculated by taking the difference of bond dissociation energies ( $\Delta\text{BDEs}$ ) between  $\text{X}-\text{O}$  and  $\text{X}-\text{H}$ , where  $\text{X}$  is either a boron or silicon atom.<sup>36</sup>  $\Delta\text{BDE}$  is found to be  $\sim 114$  kcal/mol for boranes and  $\sim 119$  kcal/mol for silanes.<sup>40</sup> These similar  $\Delta\text{BDE}$  values explain the similarity in the ranges of  $s_N$  for boranes and silanes observed in Table 1.

We now discuss our predictions for the nucleophilicity  $N$  values for each hydride donor class in detail, beginning with the carbon-based hydrides. All of the graphs presented in the following section have been generated via the  $T\Delta S^\ddagger$  correction method proposed by Morokuma and co-workers;<sup>34</sup> for the results of the other method, see section B of the Supporting Information.

**3.3. Carbon-Based Hydrides.** The carbon-based family of hydrides is of central importance and interest because they exhibit catalytic behavior and are thus recyclable. As mentioned above and to be discussed further, carbon-based hydride donors regain their aromatic nature following HT, whereas various other hydride classes cannot be easily reconverted back into hydrides. Scheme 1 illustrates the general reaction pathway of the HT reaction within the carbon-based class, which is of type (a) in Figure 1.

The aromatic catalyst resting state from which carbon-based hydrides are derived contains one or more six-member rings with conjugated  $\pi$ -bonds and  $4n + 2$  electrons in the  $\pi$ -space, thus satisfying the Hückel criteria for aromaticity. A hydridic hydrogen of the hydride is directly bonded to a ring carbon atom. We will examine here aromatic heterocycles where heteroatoms substitute for carbon atoms of the ring and maintain aromaticity as well as hydrides based on aromatics where functional groups such as alkanes and halogens are substituted on the aromatic rings' periphery.

As alluded to above, rearomatization is the major driving force for HT from such structures.<sup>12</sup> For instance, rearomatization drives HT from NADH of the analogous NADH/ $\text{NADH}^+$  redox couple active in biological reductions,<sup>41</sup> and from pyridinic analogues of NADH that effect other reductions in various biological systems. In addition, dihydropyridine—a hydride derived from pyridine—has been proposed to reduce  $\text{CO}_2$  to methanol in certain photocatalytic systems.<sup>12,20</sup> All of these carbon-based hydrides can span a large scale of nucleophilicity values. We have omitted molecules with negative  $N$  values because they are insufficiently “hydridic” to effect HT to  $\text{CO}_2$  and thus not of interest here.

Carbon-based hydrides exhibit a large  $N$  value range compared to the other three classes examined. Figure 2 displays our fit via eq 3 of experimental  $N$  values against the computed  $\Delta G^\ddagger$  for carbon-based hydrides. The  $s_N$  values found range from 0.8 to 1.1. Despite this departure from our assumption that  $s_N$  is constant for a given hydride class, the carbon-hydride Figure 2 indicates that the resulting trend enables the recovery of  $N$  values relatively accurately from the hydride donor HT to  $\text{CO}_2$  activation free energies. Linear regression of the dependence of the nucleophilicity on the activation barrier produces the equation  $N = -0.51\Delta G^\ddagger + 20.2$ .

**3.4. Benzimidazole-Based Hydrides.** The other organic hydride class considered consists of benzimidazole-based hydrides, which are generally more nucleophilic than carbon-based and silicon hydrides but are less nucleophilic than boron hydrides. Their reactions are examined here in dichloromethane solvent. Even though Mayr and co-workers determined  $N$  values of such hydrides in acetonitrile solvent,

they indicate that their  $N$  values should be similar in both solvents because the charges on both reactant species are maintained during the TS, and thus are not significantly affected by solvent polarity.

The reaction mechanism for this class is illustrated in Scheme 2 and is the second example of the type (a) reaction shown in Figure 1. A geometry optimization of the benzimidazole product molecule in acetonitrile indicates that the *cis* configuration of the two methyls attached to the two nitrogen atoms of the five-member ring is the thermodynamically stable configuration (see Scheme 2). As the HT transition state is approached, the methyl groups rotate to a planar configuration; upon HT to CO<sub>2</sub>, the positive charge produced on the oxidized hydride donor localizes on the nitrogen atoms and not the carbon originally bound to the hydride (C<sub>2</sub> carbon). The benzimidazole molecule is stabilized by resonating  $\pi$ -bonds between nitrogen atoms and the C<sub>2</sub> carbon atom, as shown in Scheme 2.

We limit our study of this hydride class to the small set of benzimidazole-based hydrides experimentally examined by Mayr and co-workers.<sup>17</sup> Figure 3 shows that the plot of the benzimidazole-based hydrides' nucleophilicity values versus the corresponding activation energy of HT to CO<sub>2</sub> exhibits an approximately linear trend, although the deviations appear large due to the figure's zoomed-in scale. Linear regression results in the equation  $N = -0.63\Delta G^\ddagger + 21.4$ .

**3.5. Boron Hydrides.** Boron hydrides are a class of hydrides commonly employed for chemical reductions.<sup>39</sup> The feature that the hydride is bound to boron significantly affects its reactivity compared to the carbon-based hydrides: boron hydrides are generally more nucleophilic than carbon-based (and silicon) hydrides.<sup>17</sup>

Most of these hydrides have a structure similar to ammonia borane, where the boron atom bound to the hydride is also bound to a nitrogen atom through a dative-covalent bond.<sup>7,43</sup> Upon HT to CO<sub>2</sub>, whose mechanism is illustrated in Scheme 3, a positive charge localizes on the boron atom, making the cation unstable and a strong Lewis acid. The negatively charged oxygens of formate act as Lewis bases, with one oxygen atom dative bonding to the boron to stabilize the oxidized product. The Scheme 3 reaction is of type (b) shown in Figure 1.

Although the  $s_N$  values of the boron hydrides considered range from 0.67 to 0.81, the nucleophilicity values still depend nearly linearly on the activation energy of HT where linear regression produces the linear relationship  $N = -0.28\Delta G^\ddagger + 17.2$ , as shown in Figure 4.

**3.6. Silicon Hydrides.** The final hydride class we consider is silicon hydrides. Their general structure involves a tetrahedral Si atom bound to a hydride and three functional groups. These hydrides are less nucleophilic than carbon and boron hydrides.<sup>39</sup>

As illustrated in Scheme 4, the reaction pathway of a HT from silicon hydrides to CO<sub>2</sub> is analogous to that of boron hydrides and is another example of reaction type (b) shown in Figure 1.

A positive charge localizes on the silicon atom upon the transfer of the hydridic H from the silicon hydride to CO<sub>2</sub>. The negatively charged oxygen of formate forms a covalent bond with silicon, as illustrated in Figure 5. The formation of this strong and stable Si–O bond provides the driving force for the hydride transfer.

The nucleophilicity values of silicon-based hydrides were regressed on the activation energies of hydride transfer to CO<sub>2</sub>,

and the linear correlation  $N = -0.68\Delta G^\ddagger + 23.9$  shown in Figure 6 was obtained. Although the sensitivity parameter varies from 0.58 to 0.79 for silicon hydrides, the accuracy of the model of eq 3 with constant  $s_N$  has been established on the basis of the 95% confidence interval.

**3.7. Bell–Evans–Polanyi Relationships.** Consistent with the Bell–Evans–Polanyi principle, we find that  $N$ —or equivalently, the activation free energy—depends linearly on  $\Delta G_{\text{rxn}}$  of the hydride transfer reaction for each hydride donor class, as shown in Figure 7.<sup>44,45</sup> This is especially helpful because calculations of the thermodynamic reaction free energies only require computing the reactant and product free energies, which is considerably simpler than calculating TS free energies. Our result is analogous to recent efforts by Allgäuer et al., who relate reaction energies to the electrophilicities of various Michael acceptors.<sup>46</sup> However, directly relating  $\Delta G^\ddagger$  to  $N$  yields slightly more accurate  $N$  values. The boron-based hydrides are an exception where the  $N$  values are considerably less accurate, a feature likely due to the two outliers lying significantly off the linear fit.

**3.8. General Trends.** We now attempt to clarify some trends exhibited by the derived linear models of each hydride transfer class. Effective  $s_N$  values were determined by simple algebraic manipulations on the slope of each line, as presented in eq 3. The  $s_N$  values recovered using our method lie above the respective ranges of those determined by Mayr and co-workers, as given in Table 1. While this issue of  $s_N$  values indicates a certain qualification necessary for our method, it still achieves the main goal of correctly predicting  $N$  values. We consider that the disparity in the prediction of effective  $s_N$  values compared to those obtained by the Mayr group arises from the difference between our analysis and that of that group. Our approach requires the consideration of the single electrophile CO<sub>2</sub>, while  $s_N$  was deduced in the Mayr group's work from experiments of a single nucleophile with many electrophiles.<sup>17</sup> (This is the qualification that we mentioned above.) It is important to stress that this difference between our predicted  $s_N$  value and the Mayr group's  $s_N$  values does not change our physical interpretation of  $s_N$ , as discussed in subsection 3.2. If one is interested in determining the  $s_N$  value of a new hydride molecule (e.g., to estimate the rate constant  $k$  using eq 1), we recommend using an effective  $s_N$  that is the average of the  $s_N$  values spanned by the class to which it belongs.

We find that each hydride class exhibits a distinct slope in the  $N$  vs  $\Delta G^\ddagger$  relation, a behavior consistent with our qualitative interpretation of the correlation of  $s_N$  values with the reaction product stabilization, but this behavior indicates that a universal model to predict the nucleophilicities of all organic hydrides cannot be established using our method. It is thus not surprising that combining the data for all four hydride classes neither provides further insight nor aids in determining nucleophilicity values, as the data is very poorly fit with a MAD of 1.06. Nonetheless, it is important to stress that—as further supported in the Concluding Remarks—our approach does achieve the goal of predicting the strength of various hydrides by accurately approximating their nucleophilicity values, as shown in Figure 8. We will collect our results for molecules with known experimental  $N$  values in the next section, but here we illustrate our main goal in Table 2 by giving several examples of predicted  $N$  values for new molecules, using the linear relationships established in Figures 2, 3, 4, and 6.

Table 3. Summary of Experimental and Recovered  $N$  Values<sup>a</sup>

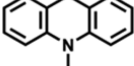
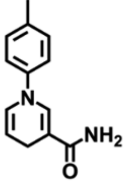
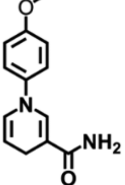
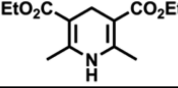
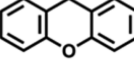
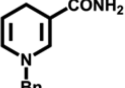
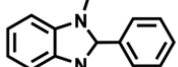
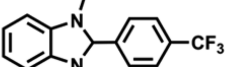
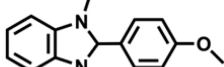
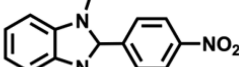
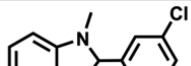
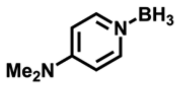
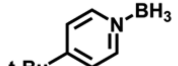
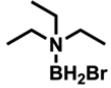
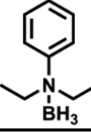
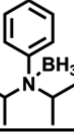
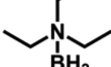
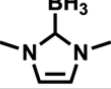
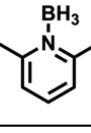
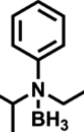
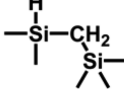
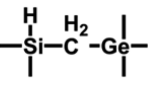
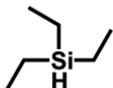
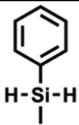
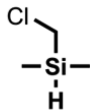
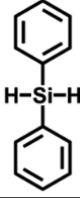
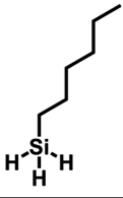
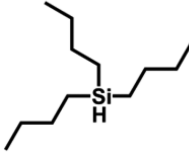
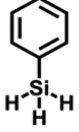
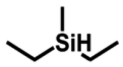
| Molecule                                                                            | $N_{\text{exp}}$ | $N_{\text{pred}}$ | $\Delta N$ | Ratio | Molecule                                                                             | $N_{\text{exp}}$ | $N_{\text{pred}}$ | $\Delta N$ | Ratio |
|-------------------------------------------------------------------------------------|------------------|-------------------|------------|-------|--------------------------------------------------------------------------------------|------------------|-------------------|------------|-------|
|    | 0.09             | 0.62              | -0.53      | 6.85  |    | 5.54             | 4.73              | 0.81       | 0.85  |
|    | 7.68             | 7.61              | 0.07       | 0.99  |    | 8.11             | 7.54              | 0.57       | 0.93  |
|    | 9                | 8.54              | 0.46       | 0.95  |    | 0.64             | 0.72              | -0.08      | 1.12  |
|    | 7.53             | 7.79              | -0.26      | 1.03  |    | 8.67             | 9.72              | -1.05      | 1.15  |
|    | 9.72             | 9.81              | -0.09      | 1.01  |    | 8.74             | 8.62              | 0.12       | 0.99  |
|    | 10.01            | 9.95              | 0.06       | 0.99  |    | 8.36             | 8.47              | -0.11      | 1.01  |
|  | 9.38             | 9.36              | 0.02       | 1.00  |                                                                                      |                  |                   |            |       |
|  | 11.01            | 11.11             | -0.10      | 1.01  |  | 12.44            | 12.32             | 0.12       | 0.99  |
|  | 10.46            | 10.38             | 0.08       | 0.99  |   | 10.01            | 10.23             | -0.22      | 1.02  |
|  | 7.97             | 8.28              | -0.31      | 1.04  |  | 7.49             | 7.74              | -0.25      | 1.03  |
|  | 8.53             | 7.95              | 0.58       | 0.93  |  | 8.84             | 8.81              | 0.03       | 1.00  |
|  | 8.9              | 9.23              | -0.33      | 1.04  |  | 11.88            | 11.60             | 0.28       | 0.98  |
|  | 10.33            | 10.62             | -0.30      | 1.03  |  | 9.12             | 8.71              | 0.41       | 0.96  |
|  | 4.86             | 4.11              | 0.75       | 0.85  |  | 5.36             | 4.98              | 0.38       | 0.93  |

Table 3. continued

| Molecule                                                                          | $N_{\text{exp}}$ | $N_{\text{pred}}$ | $\Delta N$ | Ratio | Molecule                                                                           | $N_{\text{exp}}$ | $N_{\text{pred}}$ | $\Delta N$ | Ratio |
|-----------------------------------------------------------------------------------|------------------|-------------------|------------|-------|------------------------------------------------------------------------------------|------------------|-------------------|------------|-------|
|  | 3.58             | 4.30              | -0.72      | 1.20  |  | 2.13             | 2.11              | 0.02       | 0.99  |
|  | 0.79             | 0.73              | 0.06       | 0.92  |  | 3.15             | 3.27              | -0.12      | 1.04  |
|  | 1.52             | 1.53              | -0.01      | 1.01  |  | 0.19             | 0.03              | 0.16       | 0.14  |
|  | 3.99             | 4.24              | -0.25      | 1.06  |  | 0.06             | 0.22              | -0.16      | 3.64  |
|  | 3.3              | 3.57              | -0.27      | 1.08  |  | 3.4              | 3.24              | 0.16       | 0.95  |

<sup>a</sup>See Table S8 in the Supporting Information for  $N$  values predicted using approach 2.

**3.9. Catalytic Applicability.** Hydrides must possess several properties to render them competent catalysts for the reduction of  $\text{CO}_2$  to formate and, eventually, to methanol. They must be sufficiently strong hydrides, i.e., possess favorable thermodynamics, to successfully transfer the hydride to  $\text{CO}_2$ . Furthermore, the HT should occur at reasonable rates. For instance, the activation barriers should be less than  $\sim 20$  kcal/mol at 298 K for practical HT rates. Analogously, the  $N$  parameter, which is directly related to the activation barrier, allows comparisons to be made between well-studied catalysts, such as dihydropyridine, and other prospective catalysts. Lastly, the active hydride species must be easily regenerated. The technicalities of hydride regeneration are outside the scope of this work and not discussed in detail. However, we note that, although boron- and silicon-based hydrides satisfy the thermodynamic criteria, hydride regeneration is inhibited by the formation of stable B–O and Si–O bonds. As such, they do not act catalytically in reducing  $\text{CO}_2$ . Carbon-based hydrides span a wide range of thermodynamic and kinetic scales. They are relatively weak compared to silicon- and boron-based hydrides. However, the stronger carbon-based pyridinic and benzimidazole-based hydrides are worth investigating as  $\text{CO}_2$  reduction catalysts. Although they have unfavorable  $\Delta G_{\text{rxn}}$  and large  $\Delta G^\ddagger$  in DCM, both the thermodynamics and kinetics of HT are significantly aided in aqueous media due to substantial stabilization of the hydride and formate ions by the polar solvent.

#### 4. CONCLUDING REMARKS

The success of our method for employing quantum chemically calculated hydride transfer activation energies to predict kinetic nucleophilicity values  $N$  is illustrated in Figure 8. Good agreement between the recovered and experimental  $N$  values is

obtained for all of the hydride donors to  $\text{CO}_2$  considered in the present work. Table 3 on the following page tabulates the recovered  $N$  values and compares them to experimental data.<sup>17</sup> The MAD of the difference between experimental and predicted  $N$  values of all examined species is 0.19. Only one molecule deviated by more than 1 nucleophilicity unit. Of the 37 molecules examined, 4 deviated from the experimental value by more than 20%. A total of 28 out of 37 hydride donors'  $N$  values were predicted to within 10% of the experimental value. Encouraged by this success, we have indicated in Table 2 some predicted  $N$  values for future experiments on some hydride donors in the four classes examined in this work.

We have selected  $\text{CO}_2$  as the reference hydride acceptor in this work for the reasons indicated in the Introduction, but different hydride acceptors could be chosen. While the resulting slope and intercept values will likely differ with a different reference acceptor, we anticipate that the method will be equally accurate for computationally screening promising hydrides. We propose that the model may also be used to predict electrophilicity values at constant  $N$ .

In summary, we have demonstrated the ability of our method to accurately approximate the kinetic nucleophilicity of various organic and inorganic hydride donors using computational methods. In particular, we show that, when selecting the same hydride acceptor (here  $\text{CO}_2$ ) as a reference, the hydride donor's kinetic nucleophilicity is linearly correlated with the free energy barrier of hydride transfer for each class of hydrides. Each class exhibits a unique linear relationship of nucleophilicity  $N$  with the hydride transfer activation barrier  $\Delta G^\ddagger$ , with the correlations' uniqueness stemming from each class's reactive nature. This method will enable the rapid screening of hydride donors in the search for organic hydrides with the ability to catalytically reduce various oxidants. A specific

reduction of central interest is that of CO<sub>2</sub> to formate. Earlier efforts in this group have considered 1,2-dihydropyridine in this connection.<sup>12,20</sup> In future work, we will use the models developed in the present work to study the effects on the nucleophilicity of functionalizing dihydropyridines with various groups in order to aid the design of CO<sub>2</sub> reduction catalysts.

## ■ ASSOCIATED CONTENT

### ● Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcb.7b12093.

Computational methods; detailed derivation of eq 4; benchmarking calculations; results using alternative corrections for activation entropies; molecular coordinates and zero-point energies for all hydrides and corresponding transition states (PDF)

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### Notes

The authors declare no competing financial interest.

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